```
!pip install transformers
!pip install datasets
!pip install scikit-learn
!pip install pandas
!pip install matplotlib
!pip install seaborn
!pip install torch
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.25.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.6.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.3.2)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (4.15.0)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)
Requirement already satisfied: certifi in
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/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (2026.2.25)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0-
>transformers) (3.11)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub<2.0,>=1.3.0->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub<2.0,>=1.3.0->transformers) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers)
(0.1.2)
Requirement already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.25.1)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in
/usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
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Requirement already satisfied: xxhash in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: huggingface-hub>=0.24.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (1.6.0)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.3)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0-
>datasets) (1.3.2)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0-
>datasets) (0.28.1)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0-
>datasets) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0-
>datasets) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.4.5)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2026.2.25)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2025.3)
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Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.4.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.23.0)
Requirement already satisfied: anyio in
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>huggingface-hub>=0.24.0->datasets) (4.12.1)
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/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.24.0->datasets) (1.0.9)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets) (0.16.0)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-
>pandas->datasets) (1.17.0)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub>=0.24.0->datasets) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub>=0.24.0->datasets) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub>=0.24.0->datasets) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub>=0.24.0->datasets) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub>=0.24.0->datasets) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
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/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub>=0.24.0->datasets) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer->huggingface-hub>=0.24.0->datasets) (0.1.2)
Requirement already satisfied: scikit-learn in
/usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-
>pandas) (1.17.0)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: numpy>=1.23 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib)
(2.9.0.post0)
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Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7-
>matplotlib) (1.17.0)
Requirement already satisfied: seaborn in
/usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in
/usr/local/lib/python3.12/dist-packages (from seaborn) (2.0.2)
Requirement already satisfied: pandas>=1.2 in
/usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in
/usr/local/lib/python3.12/dist-packages (from seaborn) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (4.62.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (26.0)
Requirement already satisfied: pillow>=8 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn)
(2025.3)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7-
>matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Requirement already satisfied: torch in
/usr/local/lib/python3.12/dist-packages (2.10.0+cu128)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from torch) (3.25.1)
Requirement already satisfied: typing-extensions>=4.10.0 in
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/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: cuda-bindings==12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
/usr/local/lib/python3.12/dist-packages (from torch) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4-
>torch) (1.4.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch)
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(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
```

```python
import pandas as pd

fake = pd.read_csv("Fake.csv")
true = pd.read_csv("True.csv")

fake["label"] = 1
true["label"] = 0
df = pd.concat([fake, true])
df["content"] = df["title"] + " " + df["text"]
df = df[["content","label"]]
df
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never paid much attention to politics in the past. She plans to cast her first ever vote for Trump after hearing his proposed policies to curb illegal immigration, which include building a giant wall on the southern border and making Mexico pay for it. \\u201cWe need a piranha in there, and he\\u2019s definitely a piranha,\\u201d she said. Trump and his operatives are confident they can do what few of his rivals for the Republican nomination have shown they can do \\u2014 expand the party\\u2019s potential voter pool. But while Democrat Barack Obama did that eight years ago by largely registering new voters, including record levels of minorities, both male and female, in urban centers, Trump\\u2019s campaign has instead largely sought out the disaffected, who tend to be overwhelmingly older, white, and less educated than the broader electorate. \\u201cMy whole campaign has been focused on expanding the number of people who want to, and will, participate in this election cycle,\\u201d Trump wrote in a recent op-ed in USA Today. When analyzing \\u201clost\\u201d or \\u201cnew\\u201d voters, Reuters compiled poll results from people who haven\\u2019t voted in the previous two presidential campaigns and midterm congressional elections. The results focused only on those who said they were nearly certain to vote in the November election. It included responses from 3,440 \\u201cnew\\u201d Republican and independent voters \\u2013 a sample that has a credibility interval of 2 percentage points. Jan Leighley, an expert on turnout at American University, said it\\u2019s too soon to compare Trump\\u2019s \\u201cnew\\u201d voters with the disenfranchised voters, especially minorities, who in 2008 turned out in record numbers to elect the first black president. Those voters didn\\u2019t find their way to the polls simply because they were inspired to make history, Leighley said. \\u201cHe also had a kick-ass mobilization structure.\\u201d \\u201cObama\\u2019s campaign went door-to-door. They canvassed whole neighborhoods. \\u201cI don\\u2019t know if he (Trump) has the campaign infrastructure to make sure that the folks who are riled up are walked to the polls on Election Day,\\u201d she said. Trump\\u2019s campaign insists he does have the infrastructure, but won\\u2019t provide specifics on how it intends to turn casual supporters into engaged voters. The campaign says it does follow up with the thousands of attendees who jam arenas for Trump\\u2019s rallies. That\\u2019s one significant advantage the reality star has over the other candidates in the Republican field, as the challenge in reaching voters who have fallen out of the political process often lies in simply locating them. \\u201cWe\\u2019ve identified a lot of people in early primary states who have not participated in the process before,\\u201d campaign manager Corey Lewandowski told Reuters. Lewandowski resists the notion that Trump needs to turn out \\u201cnew\\u201d or reluctant voters to win states such as Iowa. Even if the Trump campaign isn\\u2019t necessarily counting on them, it may be reassuring that his supporters among \\u201cnew\\u201d voters who responded to the Reuters/Ipsos survey appear motivated to cast a vote for him. At least for the time being. Ronald Thomas, a 49-year-old

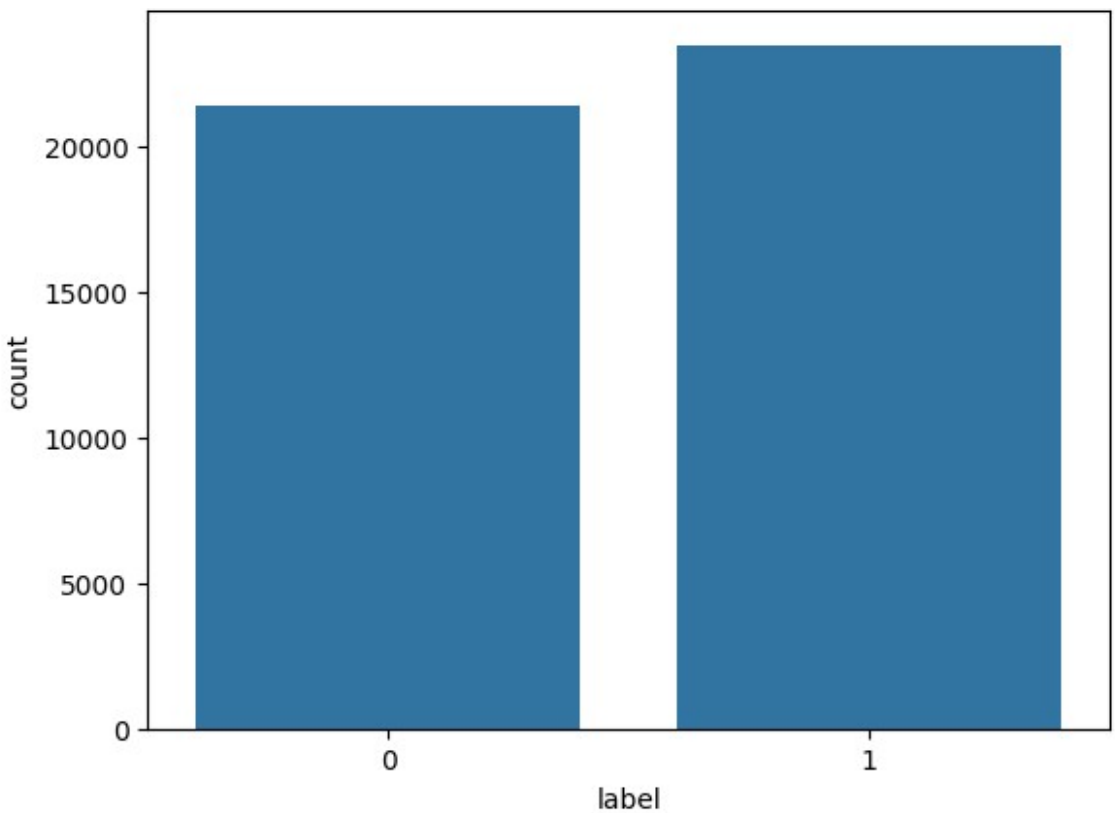
truck driver and Navy veteran in North Carolina, said he, too, has never voted in a presidential contest. His girlfriend would push him to vote but he would always say \\u201cYeah, but the right one ain\\u2019t come along yet,\\u2019\\u201d Thomas said. Trump is that man. His willingness to take on the government has set him apart as someone who would \\u201cactually look out for the people,\\u201d said Thomas. Now he wants to know how to register so that he can vote for the billionaire. And there\\u2019s Vince DiSylvester, a retired maintenance worker in Missouri, who, at 73, said has never cast a ballot for president. But Trump has inspired him to rethink that. \\u201cHe\\u2019s a businessman, he knows business, he knows how to get things done,\\u201d he said. \\u201cAnd he tells it like it is. If you don\\u2019t like it - well, too bad.\\u201d (Editing by Ross Colvin) SAP is the sponsor of this coverage which is independently produced by the staff of Reuters News Agency.\",\n   \"Iranian Revolutionary Guards commander, another fighter, killed in Syria: Iranian media BEIRUT (Reuters) - A commander in Iran s elite Revolutionary Guards and a lower-ranking Iranian fighter have been killed fighting Islamic State in Syria in recent days, Iranian media reported on Sunday. The Revolutionary Guards, Iran s most powerful military force which also oversees an economic empire worth billions of dollars, have been fighting in support of Syrian president Bashar al-Assad for several years. An Iranian official told the Tasnim news agency last year that more than 1,000 Iranians have been killed in Syria. Senior members of the Guards have been among those killed. Kheyrollah Samadi, a Guards commander in charge of a unit in Syria, died on Thursday in fighting in the Albu Kamal region, bordering Iraq, according to Fars News. Samadi was killed in clashes with Islamic State, according to the Ghatreh news site. Iranian media have previously reported on fighting in that area between Iran s Shi ite militia allies and Islamic State. The Syrian army and its allies took complete control over Albu Kamal, Islamic State s last significant town in Syria, a military news service run by Hezbollah said on Sunday. Samadi, who fought in the Iran-Iraq war during the 1980s and had retired from the Iranian military before signing on to go to Syria, was killed by a mortar explosion, Fars News, a news agency, said. Iranian news sites posted pictures on Sunday of Samadi with Qassem Soleimani, head of the Guards branch responsible for operations outside Iran. The lower-ranking Iranian fighter, Mehdi Movahednia, was killed on Saturday in clashes with Islamic State in the town of Mayadin in eastern Syria, Fars News reported. The Revolutionary Guards initially kept quiet about their role in the Syria conflict. But in recent years, as casualties have mounted, they have been more outspoken about their engagement, framing it as an existential struggle against the Sunni Muslim fighters of Islamic State who see Shi ites, the majority of Iran s population, as apostates. On web sites linked to the Guards, members of the organization killed in Syria and Iraq are praised as protectors of Shi ite holy sites and labeled defenders of the shrine . U.S. President Donald Trump last

month gave the U.S. Treasury Department authority to impose economic sanctions on members of the Iranian Revolutionary Guard in response to what Washington calls its efforts to destabilize and undermine its opponents in the Middle East. \",\n   \"Cambodian leader gets China's backing as West condemns crackdown PHNOM PENH (Reuters) - Cambodian Prime Minister Hun Sen won words of support from China on Monday after the United States and European Union condemned the arrest of his main rival and a widening crackdown on his critics before next year s election. A day after Kem Sokha was arrested in a midnight raid on his house, one of his deputies said donor countries should open their eyes to Cambodia s false democracy and put more pressure on Hun Sen. When asked about Kem Sokha s arrest at a press briefing in Beijing, Chinese foreign ministry spokesman Geng Shuang said China supports the Cambodian government s efforts to protect national security and stability. Opposition politicians, rights groups and independent media have come under growing pressure as next year s election approaches. It could represent Hun Sen s greatest electoral challenge in more than three decades in power. One of China s closest allies in the region, Hun Sen has increasingly ignored criticism from Western donors, whose budget support is no longer as critical as it during the early years of his rule, when Cambodia was little more than a failed state. We cannot allow foreigners to use Khmers to kill Khmers any more, Hun Sen said on Monday, referring to the Khmer Rouge genocide that destroyed Cambodia in the 1970s. Hun Sen, 65, is a former Khmer Rouge soldier who switched sides. Sokha was allowed to see a lawyer on Monday at his prison several hours from Phnom Penh near the border with Vietnam. I may lose freedom, but may freedom never die in Cambodia, Kem Sokha was quoted as saying in a post on Twitter that was repeated by his daughter, Monovithya Kem. The European Union called for his immediate release, based on the fact that he is meant to have parliamentary immunity, as an elected lawmaker. This arrest suggests a further effort to restrict the democratic space in Cambodia, the EU said in a statement. The U.S. State Department expressed grave concern at Sokha s arrest on charges it said appeared to be politically motivated. It said in a statement it was also worried about other curbs on media and civil society. Hun Sen has steadily increased his rhetoric against the United States, ending joint military exercises, expelling a U.S. pro-democracy group and on Sunday accusing Washington of conspiring with Kem Sokha. U.N. High Commissioner for Human Rights Zeid Ra ad Al Hussein said he was seriously concerned about the arrest, noting that it was on the basis of the video of a speech he had made in 2013 and which had been publicly available since then. One of Kem Sokha s deputies, Mu Sochua, said the opposition had done as much as it could and would not call for demonstrations because it believed in non-violence. She called on donors to help. There isn t true peace. There has always been a false democracy, said Mu Sochua, 63, who is one of three deputies to Kem Sokha in the Cambodia National Rescue Party (CNRP). The international community have been willing to close their

eyes and play along with it. Right now all the red lines have been crossed, she told Reuters in an interview in Phnom Penh. An independent newspaper that had often been critical of Hun Sen published its last edition on Monday, saying it had been forced to close after being given one month to pay a crippling $6.3 million in back taxes . Its final headline, on the arrest of Kem Sokha, was Descent Into Outright Dictatorship . \"\n   ],\n \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"label\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 0,\n   \"min\": 0,\n   \"max\": 1,\n   \"num_unique_values\": 2,\n \"samples\": [\n   0,\n   1\n   ],\n \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe","variable_name":"df"}

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='label', data=df)
plt.show()
```




```
import torch
from transformers import AutoTokenizer
from sklearn.model_selection import train_test_split
```

```
train_texts, val_texts, train_labels, val_labels = train_test_split(
    df["content"],
    df["label"],
    test_size=0.2,
    random_state=42
)

print(len(train_texts))
print(len(val_texts))
```

```
35918
8980
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(
```


```
{"model_id":"69f1519fd21c47fb934d3734210edc12","version_major":2,"vers
ion_minor":0}

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ion_minor":0}

{"model_id":"204218cc647248aea4b8f7e57de55edb","version_major":2,"vers
ion_minor":0}

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ion_minor":0}
```


```
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```

```
train_encodings = tokenizer(
    train_texts.tolist(),
    truncation=True,
    padding="max_length",
    max_length=256
)
```

```
val_encodings = tokenizer(
    val_texts.tolist(),
    truncation=True,
    padding="max_length",
    max_length=256
)

print(train_encodings['input_ids'][0])
```

```
[101, 4977, 1005, 1055, 9413, 16168, 2319, 2758, 1057, 1012, 1055,
1012, 6744, 3247, 12517, 21112, 2006, 2375, 7571, 1006, 26665, 1007,
1011, 5037, 2343, 28117, 10139, 2361, 9413, 16168, 2319, 2056, 2006,
9432, 2008, 1057, 1012, 1055, 1012, 2343, 6221, 8398, 1055, 15140,
3247, 2000, 6807, 6744, 2004, 1996, 3007, 1997, 3956, 2001, 12517,
14353, 2006, 2248, 4277, 1012, 9413, 16168, 2319, 4092, 1999, 7571,
2044, 7566, 2007, 3539, 2704, 13573, 24529, 11514, 8180, 1010, 2036,
2056, 4977, 2359, 2000, 2156, 1037, 9879, 5576, 2006, 1996, 2479,
1997, 9719, 1010, 2021, 2056, 3306, 18543, 2015, 2020, 9992, 7566,
1012, 102, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

```
class FakeNewsDataset(torch.utils.data.Dataset):

    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels

    def __getitem__(self, idx):
        item = {key: torch.tensor(val[idx]) for key, val in
self.encodings.items()}
        item["labels"] = torch.tensor(self.labels[idx],
dtype=torch.long)
        return item

    def __len__(self):
        return len(self.labels)

train_dataset = FakeNewsDataset(train_encodings,
train_labels.tolist())
val_dataset = FakeNewsDataset(val_encodings, val_labels.tolist())

sample = train_dataset[0]

for key, value in sample.items():
    print(key, value.shape)
```

```
input_ids torch.Size([256])
token_type_ids torch.Size([256])
attention_mask torch.Size([256])
labels torch.Size([])
```

```
import torch
from transformers import AutoModelForSequenceClassification
from transformers import TrainingArguments, Trainer
from sklearn.metrics import accuracy_score,
precision_recall_fscore_support

model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

{"model_id":"ef12de39e0084f73b1970f21068ae6c6","version_major":2,"version_minor":0}

{"model_id":"68e49c73c24e4a7e83fc238d85ebe77e","version_major":2,"version_minor":0}

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-
uncased
Key                       | Status     |
-------------------------+------------+-
vocab_layer_norm.bias     | UNEXPECTED |
vocab_projector.bias      | UNEXPECTED |
vocab_transform.bias      | UNEXPECTED |
vocab_layer_norm.weight   | UNEXPECTED |
vocab_transform.weight    | UNEXPECTED |
classifier.bias           | MISSING    |
pre_classifier.bias       | MISSING    |
classifier.weight         | MISSING    |
pre_classifier.weight     | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from
the checkpoint. Consider training on your downstream task.
```

```
def compute_metrics(eval_pred):

    logits, labels = eval_pred
    predictions = logits.argmax(axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels, predictions, average="binary"
    )
```

```
    acc = accuracy_score(labels, predictions)

    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }

training_args = TrainingArguments(
    output_dir="./results",

    learning_rate=3e-5,

    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,

    num_train_epochs=3,

    eval_strategy="epoch",

    save_strategy="epoch",

    logging_dir="./logs",

    load_best_model_at_end=True,

    weight_decay=0.01
)
```

`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    compute_metrics=compute_metrics
)

print(train_dataset[0])
```

```
{'input_ids': tensor([  101,  4977,  1005,  1055,  9413, 16168,  2319,
2758,  1057,  1012,
         1055,  1012,  6744,  3247, 12517, 21112,  2006,  2375,  7571,
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        26665,  1007,  1011,  5037,  2343, 28117, 10139,  2361,  9413,
16168,
         2319,  2056,  2006,  9432,  2008,  1057,  1012,  1055,  1012,
2343,
```

```
         6221,  8398,  1055, 15140,  3247,  2000,  6807,  6744,  2004,
1996,
         3007,  1997,  3956,  2001, 12517, 14353,  2006,  2248,  4277,
1012,
         9413, 16168,  2319,  4092,  1999,  7571,  2044,  7566,  2007,
3539,
         2704, 13573, 24529, 11514,  8180,  1010,  2036,  2056,  4977,
2359,
         2000,  2156,  1037,  9879,  5576,  2006,  1996,  2479,  1997,
9719,
         1010,  2021,  2056,  3306, 18543,  2015,  2020,  9992,  7566,
1012,
          102,     0,     0,     0,     0,     0,     0,     0,     0,
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0,
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0,
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0,
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0,
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0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
```

```
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
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0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
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'attention_mask': tensor([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
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        1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]), 'labels':
tensor(0)}
```

```python
print(df["content"].iloc[0])
```

```
 Donald Trump Sends Out Embarrassing New Year’s Eve Message; This is
Disturbing Donald Trump just couldn t wish all Americans a Happy New
Year and leave it at that. Instead, he had to give a shout out to his
enemies, haters and  the very dishonest fake news media.  The former
reality show star had just one job to do and he couldn t do it. As our
Country rapidly grows stronger and smarter, I want to wish all of my
friends, supporters, enemies, haters, and even the very dishonest Fake
News Media, a Happy and Healthy New Year,  President Angry Pants
tweeted.  2018 will be a great year for America! As our Country
rapidly grows stronger and smarter, I want to wish all of my friends,
```

```
supporters, enemies, haters, and even the very dishonest Fake News
Media, a Happy and Healthy New Year. 2018 will be a great year for
America!  Donald J. Trump (@realDonaldTrump) December 31, 2017Trump s
tweet went down about as welll as you d expect.What kind of president
sends a New Year s greeting like this despicable, petty, infantile
gibberish? Only Trump! His lack of decency won t even allow him to
rise above the gutter long enough to wish the American citizens a
happy new year!  Bishop Talbert Swan (@TalbertSwan) December 31,
2017no one likes you  Calvin (@calvinstowell) December 31, 2017Your
impeachment would make 2018 a great year for America, but I ll also
accept regaining control of Congress.  Miranda Yaver (@mirandayaver)
December 31, 2017Do you hear yourself talk? When you have to include
that many people that hate you you have to wonder? Why do the they all
hate me?  Alan Sandoval (@AlanSandoval13) December 31, 2017Who uses
the word Haters in a New Years wish??  Marlene (@marlene399) December
31, 2017You can t just say happy new year?  Koren pollitt
(@Korencarpenter) December 31, 2017Here s Trump s New Year s Eve tweet
from 2016.Happy New Year to all, including to my many enemies and
those who have fought me and lost so badly they just don t know what
to do. Love!  Donald J. Trump (@realDonaldTrump) December 31, 2016This
is nothing new for Trump. He s been doing this for years.Trump has
directed messages to his  enemies  and  haters  for New Year s,
Easter, Thanksgiving, and the anniversary of 9/11.
pic.twitter.com/4FPAe2KypA  Daniel Dale (@ddale8) December 31,
2017Trump s holiday tweets are clearly not presidential.How long did
he work at Hallmark before becoming President?  Steven Goodine
(@SGoodine) December 31, 2017He s always been like this . . . the only
difference is that in the last few years, his filter has been breaking
down.  Roy Schulze (@thbthttt) December 31, 2017Who, apart from a
teenager uses the term haters?  Wendy (@WendyWhistles) December 31,
2017he s a fucking 5 year old  Who Knows (@rainyday80) December 31,
2017So, to all the people who voted for this a hole thinking he would
change once he got into power, you were wrong! 70-year-old men don t
change and now he s a year older.Photo by Andrew Burton/Getty Images.
```

```
trainer.train()
```

```
<IPython.core.display.HTML object>
```


```
{"model_id":"0193e44ae1c548d1bd0b25c6a6543441","version_major":2,"vers
ion_minor":0}
```

```
{"model_id":"2e4a57ee59394ed7b20526f5b13ad266","version_major":2,"vers
ion_minor":0}
```

```
{"model_id":"f301af0218724bb6ba7b3275fa50180c","version_major":2,"vers
ion_minor":0}
```


```
There were missing keys in the checkpoint model loaded:
['distilbert.embeddings.LayerNorm.weight',
'distilbert.embeddings.LayerNorm.bias'].
```

```
There were unexpected keys in the checkpoint model loaded:
['distilbert.embeddings.LayerNorm.beta',
'distilbert.embeddings.LayerNorm.gamma'].
```

```
TrainOutput(global_step=6735, training_loss=1.148877769824271e-06,
metrics={'train_runtime': 3198.7592, 'train_samples_per_second':
33.686, 'train_steps_per_second': 2.106, 'total_flos':
7395522399565824.0, 'train_loss': 1.148877769824271e-06, 'epoch':
3.0})
```

```
trainer.save_model("fake_news_model")
tokenizer.save_pretrained("fake_news_model")
```


```
{"model_id":"7831586b5198486fb6c4b95f2d12fd33","version_major":2,"vers
ion_minor":0}
```


```
('fake_news_model/tokenizer_config.json',
'fake_news_model/tokenizer.json')
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
trainer.save_model("/content/drive/MyDrive/fake_news_model")
tokenizer.save_pretrained("/content/drive/MyDrive/fake_news_model")
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```


```
{"model_id":"d9ea9dafdea340eabc2b8b83cb71736b","version_major":2,"vers
ion_minor":0}
```


```
('/content/drive/MyDrive/fake_news_model/tokenizer_config.json',
 '/content/drive/MyDrive/fake_news_model/tokenizer.json')
```

```
trainer.evaluate()
```

```
<IPython.core.display.HTML object>
```

```
{'eval_loss': 0.0012110454263165593,
 'eval_accuracy': 0.9998886414253898,
 'eval_f1': 0.9998943475964078,
 'eval_precision': 1.0,
 'eval_recall': 0.999788717515318,
 'eval_runtime': 65.0301,
 'eval_samples_per_second': 138.09,
 'eval_steps_per_second': 8.642,
 'epoch': 3.0}
```

```
predictions = trainer.predict(val_dataset)
```

```
<IPython.core.display.HTML object>
```

```python
preds = predictions.predictions.argmax(axis=1)

from sklearn.metrics import classification_report

print(classification_report(val_labels, preds))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      4247
           1       1.00      1.00      1.00      4733

    accuracy                           1.00      8980
   macro avg       1.00      1.00      1.00      8980
weighted avg       1.00      1.00      1.00      8980
```

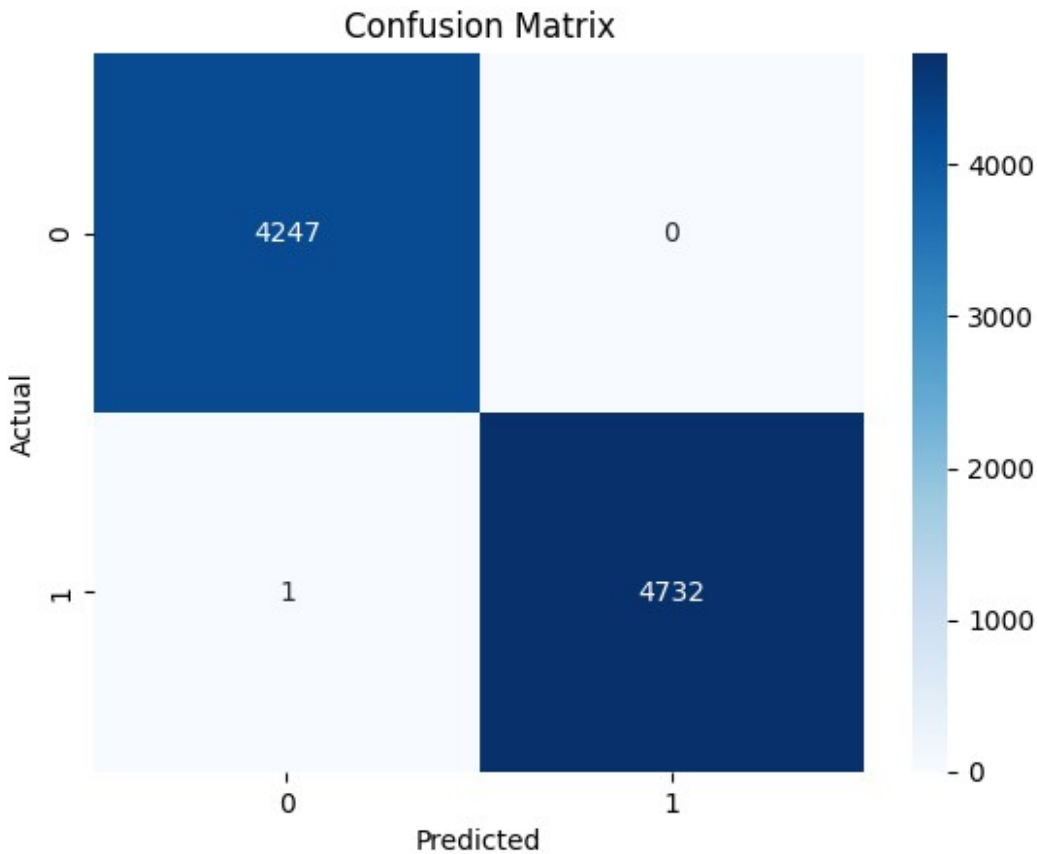
```python
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(val_labels, preds)
print(cm)
```

```
[[4247    0]
 [   1 4732]]
```

```python
import matplotlib.pyplot as plt
import seaborn as sns

sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
wrong_predictions = []

for text, true_label, pred_label in zip(val_texts, val_labels, preds):
    if true_label != pred_label:
        wrong_predictions.append((text, true_label, pred_label))

len(wrong_predictions)
```

```
1
```

```
!pip install gradio
```

```
Requirement already satisfied: gradio in
/usr/local/lib/python3.12/dist-packages (5.50.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (4.12.1)
Requirement already satisfied: brotli>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.135.1)
Requirement already satisfied: ffmpy in
/usr/local/lib/python3.12/dist-packages (from gradio) (1.0.0)
```

```
Requirement already satisfied: gradio-client==1.14.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (1.14.0)
Requirement already satisfied: groovy~=0.1 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: httpx<1.0,>=0.24.1 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.33.5 in
/usr/local/lib/python3.12/dist-packages (from gradio) (1.6.0)
Requirement already satisfied: jinja2<4.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (3.0.3)
Requirement already satisfied: numpy<3.0,>=1.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (2.0.2)
Requirement already satisfied: orjson~=3.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (3.11.7)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from gradio) (26.0)
Requirement already satisfied: pandas<3.0,>=1.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (2.2.2)
Requirement already satisfied: pillow<12.0,>=8.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (11.3.0)
Requirement already satisfied: pydantic<=2.12.3,>=2.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (2.12.3)
Requirement already satisfied: pydub in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.0.22)
Requirement already satisfied: pyyaml<7.0,>=5.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (6.0.3)
Requirement already satisfied: ruff>=0.9.3 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.15.5)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<1.0,>=0.40.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.52.1)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.24.1)
Requirement already satisfied: typing-extensions~=4.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in
/usr/local/lib/python3.12/dist-packages (from gradio) (0.41.0)
Requirement already satisfied: fsspec in
/usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0-
>gradio) (2025.3.0)
```

```
Requirement already satisfied: websockets<16.0,>=13.0 in
/usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0-
>gradio) (15.0.1)
Requirement already satisfied: idna>=2.8 in
/usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio)
(3.11)
Requirement already satisfied: typing-inspection>=0.4.2 in
/usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2-
>gradio) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2-
>gradio) (0.0.4)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1-
>gradio) (2026.2.25)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1-
>gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1.0,>=0.24.1->gradio) (0.16.0)
Requirement already satisfied: filelock>=3.10.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.33.5->gradio) (3.25.1)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.33.5->gradio) (1.3.2)
Requirement already satisfied: tqdm>=4.42.1 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.33.5->gradio) (4.67.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0-
>gradio) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0-
>gradio) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0-
>gradio) (2025.3)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2.0-
>gradio) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in
/usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2.0-
>gradio) (2.41.4)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12-
>gradio) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12-
>gradio) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12-
>gradio) (13.9.4)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-
>pandas<3.0,>=1.0->gradio) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer<1.0,>=0.12->gradio) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer<1.0,>=0.12->gradio) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer<1.0,>=0.12->gradio) (0.1.2)
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```

```
from transformers import AutoTokenizer,
AutoModelForSequenceClassification

model =
AutoModelForSequenceClassification.from_pretrained("/content/drive/
MyDrive/fake_news_model")
tokenizer =
AutoTokenizer.from_pretrained("/content/drive/MyDrive/fake_news_model"
)
```

{"model_id":"c58c15d4fd3843049921840fb3868c7b","version_major":2,"vers
ion_minor":0}

```
model.eval()
```

```
DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
```

```
          (attention): DistilBertSelfAttention(
            (q_lin): Linear(in_features=768, out_features=768,
bias=True)
            (k_lin): Linear(in_features=768, out_features=768,
bias=True)
            (v_lin): Linear(in_features=768, out_features=768,
bias=True)
            (out_lin): Linear(in_features=768, out_features=768,
bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072,
bias=True)
            (lin2): Linear(in_features=3072, out_features=768,
bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768,
bias=True)
  (classifier): Linear(in_features=768, out_features=2, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
)

import torch

def predict(text):
    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=256
    )

    outputs = model(**inputs)
    probs = torch.nn.functional.softmax(outputs.logits, dim=-1)

    pred = torch.argmax(probs).item()

    if pred == 0:
```

```
        return "Real News"
    else:
        return "Fake News"

import gradio as gr

interface = gr.Interface(
    fn=predict,
    inputs=gr.Textbox(lines=5, placeholder="Enter news text here"),
    outputs="text",
    title="Fake News Detection using DistilBERT"
)

interface.launch(share=True)
```

```
Colab notebook detected. To show errors in colab notebook, set
debug=True in launch()
* Running on public URL: https://c0f472307efb86f9dd.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU
upgrades, run `gradio deploy` from the terminal in the working
directory to deploy to Hugging Face Spaces
(https://huggingface.co/spaces)

<IPython.core.display.HTML object>
```